ENTHUSIAST, LEADER & ACHIEVER COURSE

PHASE : ALL SRG ENTHUSE, LEADER, ACHIEVER COURSE & LAKSHYA PHASE

TARGET : PRE-MEDICAL 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 17-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	2	2	2	1	4	1	1	4	2	1	2	3	2	1	3	3	3	3	3	1	1	2	1	1	1	3	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	2	2	1	4	2	2	2	3	1	1	3	2	1	3	2	2	4	4	2	4	4	1	4	1	2	1	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	2	4	4	1	1	2	4	4	3	3	3	1	1	1	3	2	1	4	2	1	3	3	4	2	2	2	4	1	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	2	4	2	4	3	1	1	3	4	2	2	1	3	1	1	4	3	2	4	2	2	2	4	2	4	1	3	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	4	1	4	3	2	1	2	1	4	2	4	4	4	3	2	3	2	2	1	3	1	1	4	3	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	4	3	2	3	2	2	1	3	2	4	1	2	2	1	1	1	2	1	2	3	3	2	1	3	4	1	2	2	4

HINT – SHEET

1. Ans (1)

$a = g - bv$

$a = 0$

$g - bv = 0 \quad \boxed{v = \frac{g}{b}}$

2. Ans (4)

Relative velocity of stone = 5 m/s

Relative acceleration of stone

$= 10 + 5 = 15 m/s^2$

$\therefore v = u + at$

$= 5 + 15 \times 2 = 35$ m/s

3. Ans (2)

$\vec{r} = \vec{r}_0(1 - \alpha t)t$

$\vec{r} = 0$ (when particle returns back to initial position)

$\vec{r} = \vec{r}_0(1 - \alpha t)t = 0$

$t = \frac{1}{\alpha}$

velocity of the particle $= \frac{dr}{dt} = r_0(1 - 2\alpha t)$

so, particle will come to rest when v = 0, i.e. after time

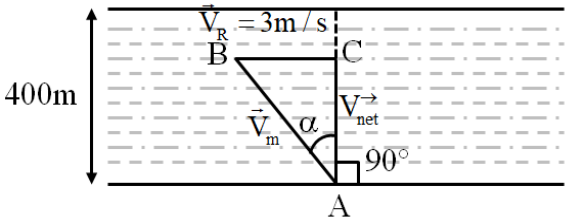
$t = \frac{1}{2\alpha}$

Position of particle at this moment

$= r_0\left[\frac{1}{2\alpha} - \alpha \times \frac{1}{(2\alpha)^2}\right]$

so, distance travel = double of the above distance $= \frac{r_0}{2\alpha}$

4. Ans (2)



Net velocity of man should be perpendicular to river flow (for zero drift)

Width of river, d = 400 m, time to cross river t = 100 sec

$\therefore |\vec{V}_{net}| = \frac{d}{t} = \frac{400}{100} = 4$ m/s

$\therefore$ In ΔABC

$\tan \propto = \frac{|\vec{V}_R|}{|\vec{V}_{net}|} = \frac{3}{4}$

$a = 37°$

$\therefore$ Total angle with downstream = 90° + 37° = 127°

5. Ans (2)

$\because x = kt \quad \therefore t = \frac{x}{k}$

$\because y = kt - akt^2 = k = \left(\frac{x}{k}\right) - \alpha k\left(\frac{x}{k}\right)^2$

$\therefore y = x - \frac{\alpha x^2}{k}$

6. Ans (1)

From given v – x graph

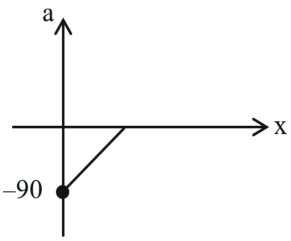
Equation is $v = -\frac{30}{10}x + 30$

$\Rightarrow v = -3x + 30$

as $a = v\frac{dv}{dx}$

$\Rightarrow a = (-3x + 30)(-3)$

$\Rightarrow a = 9x - 90$



7. Ans (4)

The value of a physical quantity remains the same regardless of the system of units, so $n_1u_1 = n_2u_2$. We need to find the numerical value n_2 in the new system. Pascal (N/m^2) is the unit for pressure, and its dimensional formula is $[ML^{-1}T^{-2}]$.

$n_1u_1 = n_2u_2$

$1\,[M_1L_1^{-1}T_1^{-2}] = n_2[M_2L_2^{-1}T_2^{-2}]$

$n_2 = \left[\frac{1}{4}\right]^1\left[\frac{1}{10}\right]^{-1}\left[\frac{1}{1}\right]^{-2} = \frac{5}{2}$

8. Ans (1)

Dimensions:

$[T] = T, [P] = ML^{-1}T^{-2}, [D] = ML^{-3}, [S] = MT^{-2}$

Assume:

$[T] = [P]^a[D]^b[S]^c = \left(ML^{-1}T^{-2}\right)^a\left(ML^{-3}\right)^b\left(MT^{-2}\right)^c$

$= M^{a+b+c}L^{-a-3b}T^{-2a-2c}$

Mass: $0 = a + b + c \Rightarrow a + b + c = 0$

Length: $0 = -a - 3b \Rightarrow a + 3b = 0 \Rightarrow a = -3b$

Time:

$1 = -2a - 2c \Rightarrow -2(-3b) - 2c = 1 \Rightarrow 6b - 2c = 1 \Rightarrow c = 3b - 0.5$

Choose $b = 1/2 \Rightarrow a = -3/2, c = 1$

$\therefore a, b, c = -\frac{3}{2}, \frac{1}{2}, 1$

9. Ans (1)

$P = mv = 3.513 \times 5.00 = 17.565$ kg m/sec.

Ans. Should be rounding to 3 S.F.

$\Rightarrow P = 17.6$ kg m/sec.

10. Ans (4)

Mean time interval

$\vec{T} = \frac{1.24 + 1.27 + 1.21 + 1.28}{4}$

$= 1.25$

Mean absolute error =

$\Delta\vec{T} = \frac{0.01 + 0.02 + 0.04 + 0.03}{4}$

$= 0.025 = 0.02$

Percentage error = $\frac{0.02}{1.25} \times 100 = 1.6\%$

11. **Ans (2)**

1. Least Count (LC) of the vernier calliper:

LC = 1 MSD − 1 VSD

Given:

- 1 MSD = 0.05 cm
- 50 VSD = 2.45 cm

So:

1 VSD=2.4550=0.049 cm

Now:

LC=0.05−0.049=0.001 cm

2. Total Reading:

Total Reading = MSR+(Vernier Coincidence X L.C)

From question:

- MSR = 5.10 cm (since zero lies between 5.10 and 5.15 cm)
- Vernier coincidence = 24

So:

Total Reading = 5.10 + (24 × 0.001)

Solution

Calculate:

24×0.001 = 0.024 cm

Total Reading = 5.10+0.024 = 5.124 cm

Correct Answer

$\boxed{5.124\text{ cm}}$

12. **Ans (1)**

If blocks move together then acceleration of system $a = \frac{24}{6} = 4\ m/s^2$

Friction force between blocks f = 2(4) = 8 N

Maximum static friction between the blocks f_{max} $= \frac{1}{2}(2)(10) = 10\ N$

As, $f < f_{max}$ Blocks move together and friction force between the two blocks will be static friction f = 8 N.

Final answer: Option (1)

13. **Ans (2)**

Let buoyant force is B.

$B - mg = \frac{mg}{6} \Rightarrow B = \frac{7mg}{6}$

S and put in box is of mass Δm then

$(m + \Delta m)g - B = (m + \Delta m)\frac{g}{6}$ …(2)

OR $(m + \Delta m)g - \frac{7gm}{6} = (m + \Delta m)\frac{g}{6}$

$m + \Delta m - \frac{7m}{6} = \frac{(m + \Delta m)}{6}$

$\Delta m - \frac{\Delta m}{6} = \frac{m}{6} + \frac{7m}{6} - m = \frac{2m}{6}$

$\Rightarrow \frac{5\Delta m}{6} = \frac{2m}{6} \Rightarrow \Delta m = \frac{2m}{5}$

Final answer : 2

14. **Ans (3)**

$F = \frac{(mV)n}{\Delta t}$

$144 = (40 \times 10^{-3})\ (1200)\ \frac{n}{\Delta t}$

$\frac{n}{\Delta t}$ = 3 bullets/sec.

15. **Ans (2)**

1. Analyze the Platform and Block: The total downward weight of the platform and the block on it is (1 kg +2 kg) × g = 3g.

2. Analyze the Upward Forces: Following the force distribution shown in the solution diagram, the total upward tension supporting the platform and block is the sum of tensions from all ropes pulling on them, which is given as 5T (where T is the tension in the single continuous rope).

3. Apply Equilibrium Condition: The total upward force must balance the total downward weight.

5T = 3g

Assuming g = 10 m/s^2, 5T = 30 N $\Longrightarrow$ T = 6 N.

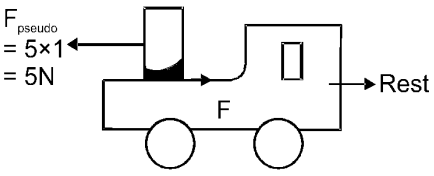
4. Analyze the Hanging Mass 'm': The hanging mass 'm' is supported by the tension T. For equilibrium:

mg = T

m(10) = 6 $\Longrightarrow$ m = 0.6 kg

16. **Ans (1)**

Solving from the frame of truck



$f \le \mu mg = 6 \Rightarrow f = 5N.$

17. **Ans (3)**

Average power,

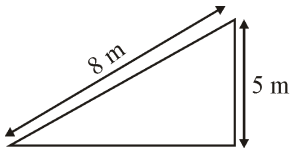
$$P_{av} = \frac{\Delta KE}{\Delta t} = \frac{\frac{1}{2}m\left(v^2 - u^2\right)}{\Delta t},$$

Here, v = 24 m/s, u = 25 m/s $\Delta t = 1s$

Therefore,

$$P_{av} = \frac{2}{2 \times 1}\left(24^2 - 25^2\right) W = -49W$$

18. **Ans (3)**



$wf_r + w_g = k_f - k_i$

$wf_r + (mg)(5) = \frac{1}{2}(m)(7)^2$

$wf_r = \frac{25 \times 49}{2} - 125g$

$-(f_r)(8) = \frac{25 \times 49}{2} - 125g$

$f_r = 76.6$ N

19. **Ans (3)**

Formula : $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$ for Conservative force

(1) $F_x = 3, F_y = 4$

$\frac{\partial F_x}{\partial y} = 0$, $\frac{\partial F_y}{\partial x} = 0$ (conservative)

(2) $F_x = 3x$, $F_y = 4y$

$\frac{\partial F_x}{\partial y} = 0$, $\frac{\partial F_y}{\partial x} = 0$ (conservative)

(3) $F_x = 3y$, $F_y = 4x$

$\frac{\partial F_x}{\partial y} = 3$, $\frac{\partial F_y}{\partial x} = 4$

(4) $F_x = 3x^2$, $F_y = 4y^2$

$\frac{\partial F_x}{\partial y} = 0$, $\frac{\partial F_y}{\partial x} = 0$ (conservative)

20. **Ans (3)**

$x = t^2 + 4t$ at $t = 0$ $v_1 = 4$ m/s

$v = (2t + 4)$ $t = 2$ $v_2 = 8$ m/s

$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$= \frac{1}{2}(2)\left[(8)^2 - (4)^2\right]$$

$= 64 - 16 = 48$ J

21. **Ans (3)**

When particle just completes the vertical circle

$V_A = \sqrt{5g\ell}$

$V_B = \sqrt{3g\ell}$

$V_C = \sqrt{g\ell}$

$\frac{K_B}{K_C} = \frac{V_B^2}{V_C^2} = 3$

22. **Ans (1)**

$a_{cp} = \omega^2 R = (20t^2)^2(0.2) = 80t^4$

$\therefore$ at $t = 0.5$ sec; $a_{cp} = 80\ (0.5)^4 = 5$ m/s^2

Now; $\alpha = \frac{d\omega}{dt} = 40t$ rad/s^2

$\Rightarrow a_t = \alpha R = 8t$ m/s^2

at $t = 0.5$ sec; $a_t = 8(0.5) = 4$m/s^2

$\therefore |\vec{a}| = \sqrt{a_t^2 + a_{cp}^2} = \sqrt{4^2 + 5^2}$

$= \sqrt{41}$m/s$^2 \simeq 6.4$m/s^2

23. **Ans (1)**

$\omega = 2\pi\ (3) = 6\pi$ rad /sec

$a_c = \omega^2 R = (6\pi)^2\ 8 = 288\ \pi^2\ \frac{cm}{s^2}$

24. **Ans (2)**

$$X_{cm} = \frac{m\left(\frac{\ell}{2}\right) + m\left(\frac{\ell}{2} + \frac{\ell}{2}\right) + m\left(\frac{\ell}{2} + \frac{\ell}{4} + \frac{\ell}{2}\right) + m\left(\frac{\ell}{2} + \frac{\ell}{4} + \frac{\ell}{6} + \frac{\ell}{2}\right)}{m + m + m + m}$$

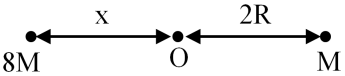
$= \frac{25}{24}\ell$

25. **Ans (1)**

let mass of a disk of radius 3R is 9M

Mass of removed hole (radius R) = M

Mass of remaining disc = 8M

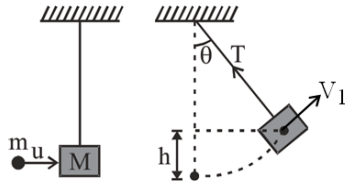


$m_1r_1 = m_2r_2$

$(8M)x = (M)(2R)$

$x = R/4$

26. Ans (1)



Applying the law of conservation of linear momentum,
$mu = (M + m)V$

$$\text{or } V = \frac{mu}{(M+m)} = \frac{0.1 \times 150}{(0.1+2.9)} = 5 \text{ m/sec}$$

$$T = (M+m)g\cos 60^\circ + \frac{(M+m)V_1^2}{\ell}$$

$$= 3 \times 10 \times \frac{1}{2} + \left(\frac{3V_1^2}{0.5}\right) = 15 + 6V_1^2 \quad ....(i)$$

According to law of conservation of mechanical energy, we have
(KE)lowest point = (KE + PE)at any other point

$$\text{or } \frac{1}{2}(m+M)V^2 = \frac{1}{2}(m+M)V_1^2 + (m+M)gh$$

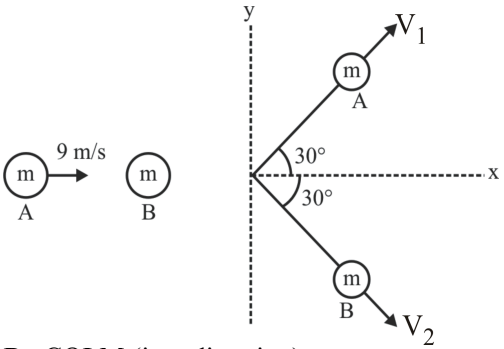
or

$$\frac{1}{2}(m+M)V^2 = \frac{1}{2}(m+M)V_1^2 + (m+M)g\ell(1-\cos\theta)$$

$$\text{or } \frac{1}{2} \times 3 \times 25 = \frac{3}{2}V_1^2 + 3 \times 10 \times 0.5\left(1 - \frac{1}{2}\right) ....(ii)$$

Solving for V_1^2, we get;
$V_1^2 = 20 \text{ (m/sec)}^2$
Putting this value in eqn. (i), we get;
$T = 15 + 6 \times 20 = 135$ N
Final Answer: Option(1) is correct.

27. Ans (1)



By COLM (in y direction) ⟶
$0 + 0 = mV_1\sin 30^\circ - mV_2\sin 30^\circ$
$V_1 = V_2$
By COLM (in x direction) ⟶
$m \times 9 + 0 = mV_1\cos 30^\circ + mV_2\cos 30^\circ$
$m \times 9 = 2mV_1\cos 30^\circ$
$V_1 = 3\sqrt{3}$ m/s
$\therefore V_1 = V_2 = 3\sqrt{3}$ m/s

28. Ans (3)

Total energy in circular orbit $E = -\frac{GMm}{2r}$

$$E_{initial} = -\frac{GM_Em}{2(2R_E)} = -\frac{GM_Em}{4R_E}$$

$$E_{final} = -\frac{GM_Em}{2(4R_E)} = -\frac{GM_Em}{8R_E}$$

Energy required $\Delta E = E_{final} - E_{initial}$

$$\Delta E = \left(-\frac{GM_Em}{8R_E}\right) - \left(-\frac{GM_Em}{4R_E}\right)$$

$$\Delta E = \frac{GM_Em}{R_E}\left(\frac{1}{4} - \frac{1}{8}\right) = \frac{GM_Em}{8R_E}$$

Final Answer: Option (3)

29. Ans (1)

$T \propto (r)^{3/2}$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2}$$

$$\frac{1 \text{ year}}{T_2} = \left(\frac{15 \times 10^{10}}{540 \times 10^{10}}\right)^{3/2} = \left(\frac{1}{36}\right)^{3/2}$$

$T_2 = 216$ years

30. Ans (3)

$$(E_K)_1 + (U)_1 = (E_K)_2 + (U)_2$$

$$\frac{1}{2}mu^2 + \left[\frac{-GM_em}{R_e}\right] = \frac{1}{2}mv^2 + \left[\frac{-GM_em}{R_e+h}\right]$$

$$\frac{1}{2}m\left(\frac{3v_e}{5}\right)^2 - \frac{GM_em}{R_e} = \frac{1}{2}m(0)^2 - \frac{GM_em}{R_e+h}$$

$$\frac{9}{50}m \times \left(\sqrt{\frac{2GM_e}{R_e}}\right)^2 - \frac{GM_em}{R_e} = -\frac{GM_em}{R_e+h}$$

$$\frac{9}{50}m \times \frac{2GM_e}{R_e} - \frac{GM_em}{R_e} = \frac{-GM_em}{R_e+h}$$

$$\frac{9GM_em}{25R_e} - \frac{GM_em}{R_e} = \frac{-GM_em}{R_e+h}$$

$$-\frac{16GM_em}{25R_e} = \frac{-GM_em}{R_e+h}$$

$$\frac{16}{25R_e} = \frac{1}{R_e+h}$$

$$h = \frac{9R_e}{16}$$

31. Ans (4)

$P \cdot E_i = -\frac{GMm}{R}$

$P \cdot E_f = -\frac{GMm}{4R}$

$\Delta P \cdot E = P \cdot E_f - P \cdot E_i$

$\Delta PE = -\frac{GMm}{4R} - \left(-\frac{GMm}{R}\right)$

$\Delta P \cdot E = \frac{3}{4}\frac{GMm}{R} \times \frac{R}{R} \quad \because \ g = \frac{GM}{R^2}$

$\Delta P \cdot E = \frac{3}{4}mgR$

32. Ans (2)

$g_{37^o} = g - R\omega^2\cos^2 37^o = g - \frac{16}{25}R\omega^2$ and

$g_{pole} = g$

$\therefore \ g_{pole} - g_{37^o} = \frac{16}{25}R\omega^2$

33. Ans (2)

Given m = 0.01 kg, g = 10m/s^2

A = 0.10m^2, h = 0.3 mm = 3×10^{-4}m, u = 0.085m/s

Tension T = mg = $0.01 \times 10 = 0.1$ N

$\eta = \frac{Fh}{A\upsilon} = \frac{0.1 \times 3 \times 10^{-4}}{0.10 \times 0.085} = 3.53 \times 10^{-3}$Pa.s

34. Ans (2)

For equillirbrium

$\vec{F}_{net} = 0$

$(\tau_{net})_{about\ any\ point} = 0$

35. Ans (1)

From Stoke's law the terminal velocity of water drop assuming buoyancy froce negligible, is

$v = \frac{2}{9}\frac{r^2\rho g}{\eta}$

$= \frac{2}{9} \times \frac{(1.5 \times 10^{-6})^2(1.0 \times 10^3) \times 9.8}{1.8 \times 10^{-5}}$

$= 2.72 \times 10^{-4}$ m/s

36. Ans (4)

$h = \frac{2T\cos\theta}{rpg}$

$\Rightarrow r = \frac{2T\cos\theta}{h\rho g}$

$= \frac{2 \times 0.1 \times \frac{1}{2}}{10 \times 10^3 \times 10} = 10^{-6}$m

37. Ans (2)

The surface tension depends upon molecular force which may different for different impurities.

Liquids form variable free surface under the molecular forces in presence of external gas pressure and the gravity.

38. Ans (2)

From Bernoulli's theorem

$P_{atm} + 0 = P_{out} + \frac{1}{2}\rho V^2$

$P_{atm} - P_{out} = \frac{1}{2}\rho V^2$

$= \frac{1}{2} \times 1.2 \times (40)^2$

$\Delta P = 960$ N/m^2

Force acting on the roof $F = (\Delta P)A$

$= 960 \times 250$

$= 2.4 \times 10^5$ N

Pressure inside the roof is greater than outside the roof, so force will act upward direction.

39. Ans (2)

$\phi\ell = \theta r$

Shear strain, $\phi = \frac{\theta r}{\ell}$

$\phi = \frac{0.8 \times 1 \times 10^{-2}}{2} = 0.004$

40. Ans (3)

$U = \frac{1}{2}kx^2 = \frac{1}{2}\left(\frac{AY}{\ell}\right)(\Delta\ell)^2 = W$

$Y = \frac{U(2\ell)}{A(\Delta\ell)^2} = \frac{2W\ell}{A(\Delta\ell)^2}$

$= \frac{2 \times 0.4 \times 1.0}{1.0 \times 10^{-2} \times 10^{-4} \times (0.2 \times 10^{-2})^2}$

$Y = 2 \times 10^{11}$ N/m^2

41. Ans (1)

$I_{total} = \frac{1}{2}(9M)R^2 = \frac{9}{2}MR^2$

Mass of removed disc $m = \frac{9M}{\pi R^2} \times \pi\left(\frac{R}{3}\right)^2 = M$

Distance of small disc center from O, $d = \frac{2R}{3}$

$I_{removed} = \frac{1}{2}M\left(\frac{R}{3}\right)^2 + M\left(\frac{2R}{3}\right)^2 = \frac{MR^2}{18} + \frac{4MR^2}{9} = \frac{MR^2}{2}$

$I_{remaining} = I_{total} - I_{removed}$

$= \frac{9}{2}MR^2 - \frac{1}{2}MR^2 = 4MR^2$

42. Ans (1)

$I = I_{cm} + md^2$

$I = \frac{2}{5}M\left(\frac{R}{2}\right)^2 + M(2R)^2 + \frac{2}{5}M\left(\frac{R}{2}\right)^2$

$I = \frac{21}{5}mR^2$

43. Ans (3)

$I = MR^2 = 8 \times (0.2)^2 = 0.32 \text{ kg} \times \text{m}^2$

$\tau = I\alpha = 0.32 \times 3 = 0.96 \text{ Nm}$

But $F = \frac{\tau}{R} = \frac{0.96}{0.2} = 4.8 \text{ N}$

44. Ans (2)

Angular Impulse = (I) (Δw)

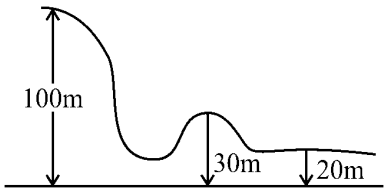
$20 = (I)(\Delta w)$

$\Rightarrow 20 = \left(\frac{2}{5}MR^2\right)(\Delta w)$

$\Rightarrow (\Delta w) = \frac{20}{(2/5)MR^2} = \frac{20 \times 5}{2 \times 2 \times (0.2)^2}$

$\Rightarrow (\Delta w) = 625 \text{ rad/s}$

45. Ans (1)



Using, energy - conservation theorem

Loss in PE = gain in kE

$mg(\Delta h) = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$\Delta h = 80\text{m}$, $\omega = \frac{v}{r}$; $I = \frac{2}{5}mr^2$; $g = 10 \text{ m/s}^2$

$v = 40\sqrt{\frac{5}{7}} \text{ m/s}.$

46. Ans (3)

Concept :-This question is based on Average Atomic weight

Solution :-

X^{40} X^{41} X^{42}

30% x 70−x

$41.25 = \frac{40 \times 30 + 41x + (70 - x)42}{100}$

$4125 = 1200 + 41x + 2940 - 42x$

$x = 15$

Final answer:- (3)

47. Ans (2)

Concept : This question is based on

Power of bulb, P = Energy emitted per second

Energy of one photon, $E = h \cdot c / \lambda$

Number of photons/sec, $n = P / E$

Solution :

Given : $P = 100$ W, $\lambda = 400\text{Å} = 4 \times 10^{-8}$ m,

$h = 6.626 \times 10^{-34}$ J·s, $c = 3 \times 10^8$ m/s

Energy of one photon:

$E = (h \cdot c)/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8) / (4 \times 10^{-8})$

$= 4.97 \times 10^{-18}$ J

Number of photons/sec:

$n = P / E = 100 / (4.97 \times 10^{-18}) \approx 2 \times 10^{19}$

Final answer : 2

48. Ans (2)

Question Explanation : We need to compare K_p and K_c. Relation is

$K_p = K_c (RT)^{\Delta n}$

If $\Delta n < 0 \longrightarrow K_p < K_c$.

Concept:- This question is based on Δn = moles of gaseous products - moles of gaseous reactants.

Solution :-

1. $N_2O_3(g) \rightleftharpoons NO_2 + NO$

$\Delta n = (1 + 1) - 1 = +1 \longrightarrow K_p > K_c$

2. $2H_2 + O_2 \rightleftharpoons 2H_2O$

$\Delta n = 2 - (2 + 1) = -1 \longrightarrow K_p < K_c$

3. $H_2 + I_2 \rightleftharpoons 2HI$

$\Delta n = 2 - 2 = 0 \longrightarrow K_p = K_c$

4. $2SO_3 \rightleftharpoons 2SO_2 + O_2$

$\Delta n = (2 + 1) - 2 = +1 \longrightarrow K_p > K_c$

Final Answer : Option (2)

$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

49. Ans (4)

Kc' (reverse) = 1 / $Kc^{1/2}$ (forward)

Solution :

Given:

- Kc for the forward reaction ($N_2 + O_2 \rightleftharpoons 2NO$) = 4×10^{-4}

Equation 1:

$N_{2(g)} + O_{(g)} = 2NO_{(g)}$

The equilibrium constant = $K_c = \dfrac{[NO]^2}{[N_2][O_2]} = 4 \times 10^{-4}$

Equation 2:

$NO \rightleftharpoons \dfrac{1}{2}N_2 + \dfrac{1}{2}O$

$K' = \dfrac{[N_2]^{\frac{1}{2}}[O_2]^{\frac{1}{2}}}{[NO]} = \dfrac{1}{\sqrt{4 \times 10^{-4}}} = \dfrac{1}{2 \times 10^{-2}} = \dfrac{100}{2} = 50$

K' = 50

Final Answer : option (4)

50. Ans (4)

For WA – SB titration P^H range will be 7 to 11,

So, an indicator of P^H range between 7 to 11 will work,

So, we will use HPh.

but MeOH has range of 3.1 to 4.4

Final answer: option (4)

51. Ans (2)

This question is based on

- 1 mole of any gas at STP occupies 22.4 L
- Moles = mass / molar mass
- Use Avogadro's number N_A to find number of atoms, electrons, or neutrons

Solution:

(A) 22.4 L CO at STP

- Volume = 1 mole $\longrightarrow$ 1 mole of CO

CO has 6(C) + 1(0) = 6 + 8 = 14 electrons (total electrons per molecule)

- number of electrons = Total electrons × N_A

$= 14 \times N_A$

- Match $\longrightarrow$ R

(B) 18 g H_2O(l)

- Mass = 18 g $\longrightarrow$ 1 mole $\longrightarrow$ contains $1 \times N_A$ molecules
- Match $\longrightarrow$ P (contains $8 \times N_A$ neutrons, as H_2O has 8 neutrons in oxygen + 0 in H)

(C) 17 g NH_3(g)

- Molar mass = 17 g/mol $\longrightarrow$ 1 mole $\longrightarrow$ contains $1 \times N_A$ molecules $\longrightarrow$ 4 atoms per molecule × N_A $\longrightarrow$ total atoms = $4 \times N_A$
- Match $\longrightarrow$ Q

(D) 10 g HF (F = 19)

- Molar mass = 1 + 19 = 20 g/mol

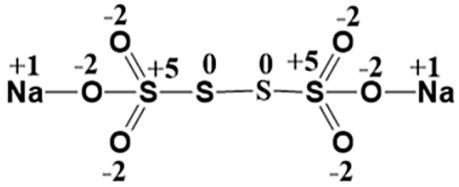
Mole = $10201020\dfrac{10}{20} \longrightarrow 0.5$ mole

- Match $\longrightarrow$ S

Final Answer: (2)

52. Ans (4)

Let the oxidation numbers of the four S atoms be x_1, x_2, x_3, x_4.



$Na_2S_4O_6 \longrightarrow 2\,Na^+$ and $S_4O_6^{2-}$

Sum of oxidation numbers in $S_4O_6^{2-}$: $x_1 + x_2 + x_3 + x_4 + 6(-2) = -2$

$x_1 + x_2 + x_3 + x_4 - 12 = -2$

$x_1 + x_2 + x_3 + x_4 = 10$

In tetrathionate, terminal S atoms are +5, and inner S atoms are 0.

Thus, oxidation numbers: +5, 0, 0, +5

Final Answer: Option 4

53. Ans (4)

$2C_{(s)} + H_{2(g)} \longrightarrow C_2H_{2(g)}\ \Delta H = \Delta H_f(C_2H_2)$

To get this equation (i) + (ii) − (iii)

$\therefore \Delta H_{f(C_2H_2)} = (-787) + (-286) - (-1301)$

$= +228$ kJ

54. Ans (1)

$W = -P_{ext}\,\Delta V$

$\Delta V = V_2 - V_1 = 13\ dm^3 - 3dm^3 = 10dm^3$

$W = -1\ atm \times 10dm^3 = -10\ atm\ dm^3$

Final Answer: Option(1)

55. Ans (4)

1. Statement 1: The azimuthal quantum number for a 'd' subshell is always 2, which is the same for 3d, 4d, and 5d. This is correct.

2. Statement 2: The d_{xy} orbital lobes are between the axes, and the p_x orbital lobes are along the x-axis. This is correct.

3. Statement 3: ψ_{520} corresponds to n = 5, 12 (d-orbital), and $m_1 = 0$ (d_{z^2}). This is correct.

4. Energies of the orbitals in the same subshell decreases with increase in atomic number.

Final answer : 4

56. Ans (1)

Sol^n / Explanation :-

• Statement I : True, The wave function (Ψ) is a function of the electron's coordinates (x, y, z) in the atom. lts value varies expending on the location within the atom.

• Statement II : False. The probability of finding an electron at a specific point is proportional to the square of the wave function (Ψ^2), not the wave function itself. Ψ^2 represents the probability density, which indicates the likelihood of finding the electron in a given region.

Therefore, statement I is true, and statement II is false

57. Ans (2)

given data Initial pressure of

$CH_3COCH_3 = 100mmHg$

$\times_{CO} = \frac{1}{3}$

Solution:

$CH_3COCH_3(g) \rightleftharpoons CH_3CH_3(g) + CO(g)$

to 100	0	0
teq 100-p	p	p
100-50	50	50

$= 50$

$P_T = 100 - \cancel{p} + \cancel{p} + p$

$= 100 + p$

$P_{co} = X_{CO} \times P_T$

$P = \frac{1}{3} \times (100 + P)$

$P = 50$

Formula $kp = \frac{P_{CH_3CH_3} \times P_{co}}{P_{CH_3COCH_3}}$

Calculation $K_p = \frac{50 \times 50}{50} = 50$

Final Answer: option (2)

58. Ans (1)

$CaCl_2 \rightleftharpoons Ca^{2+} + 2Cl^-$; if solubility = s then $[Cl^-] = 2s$ and $K_{sp} = [Ca^{2+}][Cl^-]^2$. Common-ion ($Cl^-$) from $BaCl_2$ suppresses s.

From given $[Cl^-] = 2 \times 10^{-6}$ M (pure $CaCl_2$),

$2s = 2 \times 10^{-6} \Rightarrow s = 1 \times 10^{-6}$ M.

$K_{sp} = [Ca^{2+}][Cl^-]^2 = s(2s)^2 = 4s^3 = 4 \times (1 \times 10^{-6})^3 = 4 \times 10^{-18}$.

In presence of 10^{-2} M $BaCl_2$, $[Cl^-] \approx 2 \times 10^{-2}$ M.

Let new solubility = s'.

$K_{sp} = s'([Cl^-])^2 \approx s'(2 \times 10^{-2})^2 = s' \times 4 \times 10^{-4}$.

So $s' = K_{sp} / (4 \times 10^{-4}) = (4 \times 10^{-18})/(4 \times 10^{-4})$ $= 1 \times 10^{-14}$ M.

Final answer : Option (1)

59. Ans (1)

Reactions:

$$Ag_2CO_3 \longrightarrow Ag + CO_2 + \frac{1}{2}O_2$$

$$C_2H_2 + \frac{5}{2}O_2 \longrightarrow 2CO_2 + H_2O$$

Moles $CO_2 = \frac{11.2}{22.4} = 0.5$ mol

Moles $C_2H_2 = \frac{0.5}{2} = 0.25$ mol

O_2 required $= 0.25 \times \frac{5}{2} = 0.625$ mol

From Ag_2CO_3 : 1 mol $\longrightarrow \frac{1}{2}$ mol $O_2 \Rightarrow$ required $= \frac{0.625}{0.5} = 1.25$ mol

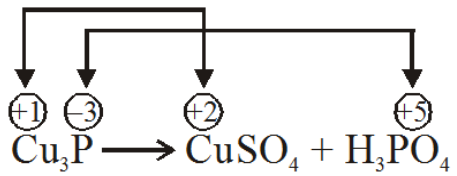
Molar mass $Ag_2CO_3 = 2 \times 108 + 12 + 48 = 276$ g

Mass pure $Ag_2CO_3 = 1.25 \times 276 = 345$ g

% purity $= \frac{345}{1380} \times 100 = 25\%$

Final answer: option (1)

60. Ans (1)



n factor of $Cu_3P = (3 \times 1) + (1 \times 8) = 11$

gram eq of Cu_3P = gram eq of $K_2Cr_2O_7$

$$(\text{Mole} \times \text{nf})_{Cu_3P} = (\text{Mole} \times \text{nf})_{K_2Cr_2O_7}$$

$1 \times 11 = \text{Mole} \times 6$

$\Rightarrow$ Mole $= \frac{11}{6}$

61. Ans (2)

Due to high hydration energy of F^-

62. Ans (2)

For spontaneous reaction $\Delta G < 0$.

63. Ans (4)

K_b (NH_4OH) $= 1.8 \times 10^{-5}$, $pK_b = -\log(1.8 \times 10^{-5})$ ≈ 4.74

K_a (CH_3COOH) $= 10^{-6}$, $pK_a = 6$

Solution:

(1) 0.01 M NH_4OH: $[OH^-] = \sqrt{(K_b \times C)}$ $= \sqrt{(1.8 \times 10^{-5} \times 10^{-2})} = \sqrt{(1.8 \times 10^{-7})}$ $\approx 4.24 \times 10^{-4}$

pOH ≈ 3.37, pH = 14 − 3.37 = 10.63 (Correct)

(2) 0.01 M CH_3COONa:

$pH = \frac{1}{2}(pK_w + pK_a + \log C)$

$= \frac{1}{2}(14 + 6 + \log 10^{-2})$

$= \frac{1}{2}(14 + 6 - 2) = \frac{1}{2}(18) = 9$ (Correct)

(3) 100 mL N/10 CH_3COOH + 100 mL N/20 NaOH

$CH_3COOH + NaOH \rightleftharpoons CH_3COONa + H_2O$

10 mmol	5 mmol	0	0
5 mmol	0	5 mmol	

$pH = pK_a + \log [Salt]/[Acid]$

Buffer with equal acid & salt: $\log [Salt]/[Acid]$ $= \log [5]/[5] = \log 1$

$\log 1 = 0$

$pH = pK_a = 6$ (Correct)

(4) 0.01 M CH_3COONH_4

$pH = \frac{1}{2}(pK_w + pK_a - pK_b)$

$pH = \frac{1}{2}(14 + 6 - 4.74) = \frac{1}{2}(15.26) \approx 7.63$

But given option says 5.53 (Incorrect)

Final Answer: Option (4)

64. Ans (4)

(A) $\overset{+3+5}{Fe}(NO_3)_3 \rightarrow \overset{+2+3}{Fe}(NO_2)_2$

Total e^- gained $= 1 + 3 \times 2 = 7$ (R)

(B) $\overset{+1-2}{CuS} \rightarrow \overset{+2}{Cu^{2+}} + \overset{+4}{SO_2}$

Total e^- lost $= 1 \times 2 + 6 \times 1 = 8$(s)

(C) $\overset{-2}{N_2}H_4 \rightarrow \overset{0}{N_2}$

Total e^- lost $= 2 \times 2 = 4$ (Q)

(D) $KI^{-1} \rightarrow \overset{+5}{K}IO_3$

Total e^- lost $= 6 \times 1 = 6$ (P)

Final answer: Option(4)

65. Ans (1)

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \text{where } n_1 = n$$

$n_2 = n + 1$

$$\therefore \quad \frac{1}{\lambda} = RZ^2\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right)$$

$$\Rightarrow \quad \frac{1}{\lambda} = \left(\frac{2n+1}{n^2(n+1)^2}\right)RZ^2$$

Since, $n >> 1$;

Therefore, $2n + 1 \approx 2n$

and $(n + 1)^2 \approx n^2$

$$\therefore \frac{1}{\lambda} = RZ^2\left(\frac{2n}{n^2.\, n^2}\right)$$

$$\Rightarrow \frac{v}{c} = \frac{2RZ^2}{n^3} \quad \text{or} \quad v = \frac{2cRZ^2}{n^3}$$

66. Ans (1)

Concept: Volume % = mole fraction for gases, Equilibrium constant (K_p)

Solution:

K_p expression depends only on gaseous species.

For the reaction: $SnO_2(s) + 2H_2(g) \rightleftharpoons 2H_2O(g) + Sn(l)$

$$K_p = \frac{(P_{H_2O})^2}{(P_{H_2})^2}$$

Mole fraction of $H_2 = 0.45$

Mole fraction of $H_2O = 0.55$

So, $P_{H_2} = 0.45P$ $P_{H_2O} = 0.55P$

$$K_p = \frac{(0.55P)^2}{(0.45P)^2}$$

$= (0.55 / 0.45)^2$

$= (11 / 9)^2$

$= 1.49$

Final Answer: Option (1)

67. Ans (2)

NCERT Pg. # 222

68. Ans (4)

Concept : This question is based on

$$\Delta S = nC_{v,m}\ln\left(\frac{T_2}{T_1}\right) + nR\ln\left(\frac{V_2}{V_1}\right)$$

Solution :

Given :

$n = 1$ mole

$$T_2 = 2T_1 \Rightarrow \frac{T_2}{T_1} = 2$$

$$V_2 = \frac{V_1}{2} \Rightarrow \frac{V_2}{V_1} = \frac{1}{2}$$

$$\Delta S = (1)\, C_{v,m}\ln(2) + (1)\, R\ln\left(\frac{1}{2}\right)$$

$\Delta S = C_{v,m}\ln(2) + R(-\ln(2))$

$\Delta S = (C_{v,m} - R)\ln(2)$

Final answer : 4

69. Ans (4)

All are correct orders

70. Ans (3)

Xe forms 4 bonds in $XeOF_2$

71. Ans (3)

CO_2 is diamagnetic

72. Ans (3)

Halogen have max EN.

73. Ans (1)

PCl_5 has axial and equitorial bonds,

BL:- Axial > equitorial (sp^3d)

74. Ans (1)

compounds having same shape

75. Ans (1)

N_2^+ and O_2^+ both have BO = 2.5.

76. Ans (3)

Hybridisation of 'B' changes from sp^2 to sp^3

77. Ans (2)

'N' does not contain vacant d-orbital.

78. Ans (1)

Net dipole moment = 0 is non polar.

79. Ans (4)

F_2 has less BDE than Cl_2

80. **Ans (2)**
Atomic no. = 114

81. **Ans (1)**
It is correct match

82. **Ans (3)**
Isoelectronic ions

83. **Ans (3)**
N > F, N < P, Si > P. since atomic size decreases from left to right & increases from top to bottom.

84. **Ans (4)**
Answer - Option(4)
Explanation - To determine the correct element based on the given first four ionization energies (19, 190, 230, and 260 eV/atom), let's analyze the ionization energy trends and understand the behavior of elements.
Concept- Given that the first ionization energy is 19 eV and the subsequent ionization energies rise moderately, the element most likely matches Sodium (Na), because sodium has a relatively low first ionization energy, and a big jump is observed after removing the second electron.
Conclusion:
The element is Sodium (Na).

85. **Ans (2)**
'O' has least EGE among gp-16.

86. **Ans (2)**
'O' has greater EN than Cl

87. **Ans (2)**
Acidic Nature $\propto$ O.S. of Non-Metal

88. **Ans (4)**
Explanation the Question : Identify the substance that can act as an acid and base.
Concept: This question is based on amphoteric substance.
Solution: Aluminum hydroxide, $Al(OH)_3$, is an amphoteric substance because it reacts with both strong acids (like HCl) and strong bases (like NaOH), demonstrating both basic and acidic properties.
SO_3, CO_2 -Acidic in nature.
Final Answer: 4

89. **Ans (1)**
Reason is correct explanation.

90. **Ans (2)**
B < C < O < N

91. **Ans (2)**
NCERT XI, Pg. # 23

92. **Ans (3)**
NCERT XI, Pg. # 7

93. **Ans (2)**
NCERT XI, Pg. # 8

94. **Ans (4)**
NCERT XI, Pg. # 10

95. **Ans (2)**
NCERT XI, Pg. # 20

96. **Ans (4)**
NCERT XI, Pg. # 17,18

97. **Ans (3)**
NCERT XI, Pg. # 20,21

98. **Ans (1)**
NCERT XI, Pg. # 90

99. **Ans (1)**
NCERT XI, Pg. # 15,16

100. **Ans (3)**
NCERT XI, Pg. # 13,14

101. **Ans (4)**
NCERT XI, Pg. # 14

102. **Ans (2)**
NCERT XI, Pg. # 17

103. **Ans (2)**
NCERT XI, Pg. # 15,90

104. **Ans (1)**
NCERT XI, Pg. # 33

105. **Ans (3)**
NCERT XI, Pg. # 29,30

106. Ans (1)
NCERT XI, Pg. # 24

107. Ans (1)
NCERT XI, Pg. # 32

108. Ans (4)
NCERT XI, Pg. # 32,33

109. Ans (3)
NCERT XI, Pg. # 26,29,32

110. Ans (2)
NCERT XI, Pg. # 25

111. Ans (4)
NCERT XI, Pg. # 32

112. Ans (2)
NCERT XI, Pg. # 27

113. Ans (2)
NCERT XI, Pg. # 32

114. Ans (2)
NCERT XI, Pg. # 99

115. Ans (4)
NCERT XI, Pg. # 98,100

116. Ans (2)
NCERT XI, Pg. # 98

117. Ans (4)
NCERT Pg. # 163
Cell after meiosis II : n and c
Cells after meiosis I : n and 2c

118. Ans (1)
NCERT XI, Pg. # 91,98,99

119. Ans (3)
NCERT XI, Pg. # 89

120. Ans (4)
NCERT XI, Pg. # 89,90,91

121. Ans (2)
NCERT XI, Pg. # 101

122. Ans (4)
NCERT XI, Pg. # 107

123. Ans (4)
NCERT XI, Pg. # 109

124. Ans (4)
NCERT XI, Pg. # 108

125. Ans (1)
NCERT XI, Pg. # 114,115

126. Ans (4)
NCERT XI, Pg. # 110, 111

127. Ans (3)
NCERT XI, Pg. # 105,108

128. Ans (2)
NCERT XI, Pg. # 125,127

129. Ans (1)
NCERT XI, Pg. # 125

130. Ans (2)
NCERT XI, Pg. # 123

131. Ans (1)
NCERT XI, Pg. # 121,122

132. Ans (4)
NCERT XI, Pg. # 125

133. Ans (2)
NCERT XI, Pg. # 121,122,124

134. Ans (4)
NCERT XI, Pg. # 123,124

135. Ans (4)
NCERT XI, Pg. # 27,29,32,33,34

136. Ans (4)
NCERT Page No. # 46

137. Ans (3)
NCERT Page No. # 40

138. Ans (2)
NCERT Page No. # 43

139. Ans (3)
NCERT Page No. # 44

140. Ans (2)
NCERT Page No. # 43

141. Ans (2)
NCERT-XII, Pg. # 54

142. Ans (1)
NCERT Page No. # 47

143. Ans (3)
NCERT, Pg. 41, 43, 44

144. Ans (1)
NCERT Pg.#46-48

145. Ans (1)
NCERT Pg.#41-45

146. Ans (4)
NCERT Page No. # 49

147. Ans (3)
Module-4

148. Ans (3)
NCERT XI Pg. # 104

149. Ans (1)
NCERT Pg. # 103

150. Ans (2)
NCERT Pg. # 103

151. Ans (3)
NCERT Pg#114

152. Ans (4)
NCERT Pg. # module

153. Ans (3)
NCERT Pg.# 80

154. Ans (2)
NCERT Pg.# 83, 84

155. Ans (3)
NCERT based

156. Ans (2)
NCERT based

157. Ans (2)
NCERT Pg. # 202, 198, 196

158. Ans (1)
NCERT Pg. # 201

159. Ans (3)
NCERT Pg. # 199, 200

160. Ans (2)
NCERT Pg. # 200

161. Ans (4)
NCERT Pg. # 195

162. Ans (1)
NCERT Pg. # 202

163. Ans (2)
NCERT Pg. # 199

164. Ans (2)
NCERT Pg. # 196

165. Ans (1)
NCERT, Pg. # 190

166. Ans (1)
NCERT, Pg. # 189

167. Ans (1)
NCERT Pg. # 185

168. Ans (2)
NCERT Pg.# 185

169. Ans (1)
NCERT Pg. 189

170. Ans (2)
NCERT Pg. No. # 186

171. Ans (3)
NCERT Pg. No. # 187, 188

172. Ans (3)
NCERT Pg. No. # 191

173. **Ans (2)**
NCERT Pg.#205,207,210, 212.

174. **Ans (1)**
NCERT Pg.#206.

175. **Ans (3)**
NCERT, Pg. # 209

176. **Ans (4)**
NCERT Page No. # 206

177. **Ans (1)**
NCERT Page No. # 208

178. **Ans (2)**
NCERT Page No. # 209

179. **Ans (2)**
NCERT, Pg. # 211

180. **Ans (4)**
NCERT Pg # 213